

Bounce-Anchored Monocular Reconstruction of Cricket Deliveries: Leg-Before-Wicket Review from a Single Phone Clip with Propagated Uncertainty

Niraj Kafle

Abstract—Ball-tracking review in cricket relies on six or more calibrated high-speed cameras and is unavailable to the club, school and net-practice cricket where the same leg-before-wicket questions arise. A single phone cannot triangulate, and a monocular projectile fit recovers depth only through the curvature gravity imposes on the image track, which is weak when the ball flies nearly along the optical axis. We show that the one event every pitched delivery contains, the bounce, removes the ambiguity. The contact instant is found in the image as the split that best explains the track with two parabolas; its pixel back-projects onto the known ground plane to a metric point with no depth cue at all; and once that anchor is fixed the remaining parameters of both parabolas are linear in the pixel measurements. A nine-parameter two-parabola model with a free post-contact horizontal velocity and an estimated restitution coefficient is refined by robust least squares, and the covariance of the fit is propagated to the stump-plane crossing so the verdict under Law 36 is widened to umpire’s call wherever the estimate’s own one-sigma bound reaches a decision boundary. On 10800 synthetic deliveries with exact ground truth, across pixel noise, frame rate, calibration error, dropped frames, unmodelled drag, swing, friction and spin, and six camera placements, the method agrees with the true verdict in 94.3% of cases with 6 false outs among 9110 not-out deliveries and a median stump-plane error of 2.7 cm laterally and 3.9 cm vertically, against 33.6% and 30.5 and 42.7 cm for the single-parabola estimator on the same detections. Through the full pipeline, with a learned detector on rendered clips, agreement is 86 of 100 with no false out. The propagated bound is slightly optimistic, covering 50.5% of lateral and 60.6% of vertical errors at one sigma, and the paper reports the real-footage behaviour, including one clip the system declines, rather than only the cases it handles.

Index Terms—Monocular 3-D reconstruction, projectile estimation, uncertainty propagation, sports video analysis, cricket, decision review, camera calibration, smartphone measurement.

I. INTRODUCTION

THE leg-before-wicket law is the most contested adjudication in cricket: an umpire must judge in real time whether a ball that struck the batter’s pad would have gone on to hit the stumps. Broadcast decision review systems answer that counterfactual by triangulating the ball from six to eight synchronised, calibrated high-speed cameras and extrapolating the post-impact path [1]–[3]. They are accurate and they are confined to televised cricket. Club, school and age-group matches, and every net session, ask the same marginal questions with no instrument at all, and it is in those settings that objective feedback has the most developmental value.

N. Kafle is an independent researcher in Kathmandu, Nepal (e-mail: contact.me.kafle@gmail.com). Source code, the synthetic harness, experiment scripts and raw result files: <https://github.com/kafle1/pocket-drs>.

This paper asks how far a single phone can go. One camera cannot triangulate, and a monocular system should not claim the accuracy of a multi-camera rig. But a cricket delivery is not an arbitrary object in an arbitrary scene. It is a sphere of known radius flying ballistically over a plane of known geometry, past two rigid regulation-height targets that are always in view, and it touches that plane exactly once. Each of those facts is a constraint, and together they are enough to make the monocular problem well posed. The paper is written as a measurement paper: a calibration from reference artefacts, an estimator with a propagated uncertainty, a test of how well that uncertainty covers the true error, and a decision rule that uses it.

Monocular projectile estimation [4] recovers scale from gravity, a cue that is strong when the parabola is seen side-on and weak when the ball travels along the optical axis, which is the natural phone position behind either set of stumps. The apparent size of the ball [5] is a second cue whose depth error grows with the square of range, and a cricket ball is a few pixels wide for most of its flight. Neither localises the ball along the pitch to better than tens of centimetres in that geometry, and Section VI shows that the resulting verdict is close to a coin toss.

Our contribution is to use the bounce as an anchor. The instant of contact is observable in the image as the split at which two parabolas explain the track far better than one. At that instant the ball’s centre is one radius above a plane whose position is known from calibration, so the contact pixel back-projects to a metric point with no depth cue at all. With the contact point and instant fixed, the velocity before and after contact are each linear in the pixel measurements, and a nine-parameter model of the whole delivery can be initialised in closed form and refined jointly. Specifically:

- We formulate delivery reconstruction as a bounce-anchored two-parabola fit with a free post-contact horizontal velocity, so seam and spin deviation off the pitch is measured rather than assumed, and an estimated restitution coefficient bounded to the physical range (Section IV).
- We propagate the covariance of the fit to the stump-plane crossing and widen the umpire’s-call band of the Laws to the estimate’s own uncertainty, so the system declines to assert a marginal call it cannot support (Section IV-G), and we measure how well the propagated bound covers the true error.
- We evaluate against exact ground truth with a generator

whose physics is richer than the estimator's (drag, lateral swing, contact friction, spin), across noise, frame rate, calibration error, dropped frames and six camera placements, and against the single-parabola estimator on identical detections; then through the full pipeline with a learned detector on rendered clips; then on real hand-held footage with a repeatability test (Section VI).

The system is built for coaching and training review, not officiating, and the paper says where it fails: a yorker leaves too few post-contact frames, a camera at the bowler's end sees the decisive arc foreshortened, and one of three real clips is declined. The system began as the author's undergraduate final-year project at Phoenix College of Management, Lincoln University College; the method, evaluation and results reported here are new.

II. RELATED WORK

A. Vision-based measurement

We treat the phone as a measuring instrument in the sense of Shirmohammadi and Ferrero [6]: the camera is the sensor, the calibration artefacts are the stumps and creases whose dimensions the Laws fix, the measurand is the ball's position at the stump plane, and the result carries an uncertainty that is itself tested. Ball trajectories have been measured with calibrated high-speed stereo rigs for table tennis [7] and golf [8], and Di Leo et al. [9] propagate covariance through stereo triangulation to an uncertainty on the reconstructed point. The present problem differs in three ways that drive the method: there is one hand-held camera whose intrinsics are unknown, the object is a few pixels wide, and the uncertainty has to be carried through a nonlinear fit and an extrapolation to a decision rather than stopping at a point.

B. Multi-camera ball tracking

Commercial review systems grew from the multi-view tracking first demonstrated for tennis [1]: intrinsic and extrinsic calibration of every camera, detection in each view, triangulation of corresponding detections [10], and a fitted flight model extrapolated to the target. Their accuracy is underwritten by wide baselines, and their cost by the installation. Collins and Evans [2] discuss how the public reads such systems' precision, and the point applies with more force to a monocular device: a verdict reported without its uncertainty invites a confidence it has not earned. Surveys of vision in sport [11] and of ball tracking [12] document the breadth of tracking work and the rarity of metric 3-D reconstruction from one view.

C. Monocular projectiles

Ribnick et al. [4] showed that the position and velocity of a projectile can be estimated from a single view by exploiting the known gravitational acceleration, and gave a linear formulation in which the six parameters of a parabola appear linearly once the projection equations are cross-multiplied. Our pre- and post-contact solves are of that form with three unknowns instead of six, because the anchor supplies the position. Chen

et al. [13] fit physics-based parabolas to basketball shots in broadcast video and Chao et al. [14] do the same from a single monocular view; Ponglertnapakorn and Suwajanakorn [15] learn to lift a 2-D ball track to 3-D across several sports. All three take scale from gravity and none reports the uncertainty of the result. Van Zandycke and De Vleeschouwer [5] regress ball depth from its apparent size in a calibrated image, which we use only in the fallback for a delivery whose contact is not observed. Closest to our setting, Ghimire et al. [16] estimate a cricket ball's trajectory from one calibrated camera: an initial depth from image-plane measurements is smoothed by a Kalman filter under a ballistic model, and the bounce is then detected on the filtered track. There the bounce is a by-product of a trajectory already estimated; here it is the anchor that makes the trajectory observable in the first place, and the verdict carries a propagated uncertainty. In racquet sports, MonoTrack [17] reconstructs shuttle trajectories from monocular badminton video and Calandre et al. [18] recover table-tennis ball kinematics from one camera; both exploit the known table or court and a bounce or hit as a constraint, in a geometry where the camera is far to the side. TNet [19] and the TrackNet family [20]–[22] address the detection of small fast balls; Yan et al. [23] the association of candidates into tracks under clutter, which our association stage follows in spirit.

D. Cricket ball flight and bounce

Mehta [24] reviews the aerodynamics of swing; Robinson and Robinson [25] give the equations of a spinning sphere under drag and lift, which our synthetic generator integrates. Carré et al. [26] model the impact of a cricket ball on a rigid surface and James et al. [27] measure the bounce and pace of real pitches, both of which inform the restitution range we allow the estimator. The leg-before-wicket law is Law 36 of the MCC Laws [28]; the umpire's-call bands used by ball tracking in the professional game are set out in the ICC playing conditions [29].

E. Estimation

Camera pose from known points follows Zhang [30] and the PnP literature [31], [32]; robust fitting under outliers follows RANSAC [33] and the robust losses of bundle adjustment [34]; the bounded trust-region solver we use is that of Branch et al. [35] as implemented in SciPy [36], and the linearised covariance of a nonlinear least-squares estimate is standard [37]. Monocular depth networks [38] recover relative depth of a static scene and structure-from-motion [39] needs camera motion; neither uses the constraint that makes this problem tractable.

III. GEOMETRY AND OBSERVABILITY

We use a right-handed pitch frame with origin at the striker's stumps: x runs down the pitch toward the bowler, y across it, z up (Fig. 1). A pinhole camera with intrinsics K and pose $(R, \mathbf{t})$ maps a world point $\mathbf{p}$ to camera coordinates $\mathbf{p}_c = R\mathbf{p} + \mathbf{t}$ and to the pixel $(u, v) = (fx_c/z_c + c_x, fy_c/z_c + c_y)$.

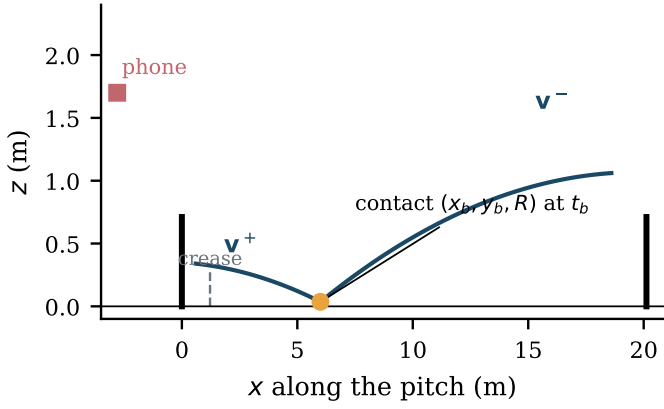


Fig. 1. The model. A delivery is two gravity parabolas joined at the contact point, whose height above the pitch is the ball radius. The phone sits behind one set of stumps, so the ball travels nearly along the optical axis and its depth is the least observable coordinate.

The stumps are $h_s = 0.711$ m high and the outer edges of a three-stump set are $\pm w_o = 0.132$ m from the middle stump; the pitch is $L = 20.12$ m long and 3.05 m wide; the ball has radius $R = 0.036$ m.

A. Why one parabola is not enough

Between release and contact the ball is a projectile, $\mathbf{p}(t) = \mathbf{p}_0 + \mathbf{v}_0 t - \frac{1}{2} g t^2 \hat{\mathbf{z}}$, with six unknowns. Each frame gives two equations, and after cross-multiplying the depth out of the projection the system is linear in the six unknowns [4]. The absolute scale is fixed only by g : a trajectory twice as far away and twice as fast projects to the same pixels at every instant except for the gravitational sag, which is the term that does not scale. Seen side-on the sag is a large fraction of the image motion; seen from behind the stumps the ball's image moves mostly along one screen axis as it approaches, and the sag is a small correction to a nearly linear image path. The linear system is then ill-conditioned along the depth-and-speed direction, and the estimate is a large distance along the pitch times a small angle.

The apparent size of the ball gives an independent depth $d = fR/r$ from its pixel radius r , but $\partial d / \partial r = -d/r$, so an error δr in the measured radius produces $\delta d = d^2 \delta r / (fR)$: quadratic in range. At 15 m with $f = 950$ px the ball is 2.3 px in radius and half a pixel of radius error is 3 m of depth. The cue seeds a fit and cannot carry one.

B. What the contact supplies

A pitched delivery touches the ground once. At the contact instant t_b the centre of the ball is at (x_b, y_b, R) : two unknowns on a known plane. If the contact pixel (u_b, v_b) can be found, the ray through it intersects the plane $z = R$ at a unique metric point, with no depth cue at all; the calibration already supplies the plane. Given that anchor, the pre-contact trajectory $\mathbf{p}(t) = (x_b, y_b, R) + \mathbf{v}^- \tau - \frac{1}{2} g \tau^2 \hat{\mathbf{z}}$, $\tau = t - t_b < 0$, has three unknowns instead of six, and the post-contact trajectory with $\mathbf{v}^+$ likewise. Each side is a linear system with two equations per frame and three unknowns, over-determined from two frames, and the

depth direction is no longer degenerate because the anchor pins the trajectory in metres at one instant and g pins the time scale. The observability gained is not incremental: it turns the ill-conditioned single-parabola solve into two well-conditioned three-parameter solves, and Section VI measures the difference as a factor of ten in error.

IV. METHOD

A. Calibration from the marks the Laws fix

The user taps the four corners of each stump set (top-left, top-right, bottom-right, bottom-left, striker's end then bowler's) and the four pitch corners. The eight stump corners lie at $(0, \pm w_o, \{0, h_s\})$ and $(L, \pm w_o, \{0, h_s\})$; the pitch corners at $(0, \pm 1.525, 0)$ and $(L, \pm 1.525, 0)$. The pose is solved by SQPnP [32], sweeping the focal length over a phone's field-of-view range and, when the caller does not pin it, the pitch length over 2 to 26 m, and keeping the lowest reprojection error. The stumps alone leave the focal-length-against-length trade-off nearly degenerate; the pitch corners, which lie off the stump plane, break it, and are dropped only when they disagree with the stumps by more than a frame-scaled tolerance, because the stump height is the one dimension the Laws fix exactly. A pose whose reprojection error exceeds $\max(25, 0.03W)$ px for frame width W , whose camera height is outside 0.1 to 5 m, or whose fitted length is outside 1.5 to 25 m is rejected with a message asking for the marks to be re-tapped. The camera's end is inferred from which stump rectangle is larger in the image, and with the batter's handedness fixes which sign of y is the leg side.

B. Detection and association

Per frame, a YOLO detector [40], [41] fine-tuned on cricket balls and a classical detector combining hue thresholding with frame-difference motion each emit candidates with a pixel centre, radius and confidence; a location detected in more than 30% of frames over more than 60% of the clip's span is static clutter and is suppressed. Candidates are associated into one track by RANSAC over a constant-acceleration image model [33]: seed pairs propose an arc, inliers within a radius scaled to the frame diagonal are counted, the best arc is refined by weighted least squares, and a second pass over the unclaimed detections stitches on the far side of a bounce when it continues the same horizontal motion. The learned track is preferred; a colour track overrides it only when it is both markedly longer and tight in an absolute sense. A sustained reversal of horizontal image travel marks the bat or pad interception and ends the live track. The output is a list of $(t_i, u_i, v_i, r_i, w_i)$: time, pixel centre, pixel radius and a confidence used as a weight. This stage is not the contribution; it is the input the reconstruction is given.

C. Finding the contact in the image

For every split index k from three frames in to three frames from the end, one quadratic in t is fitted to each pixel coordinate on each side and the total weighted residual is recorded, alongside the residual of a single quadratic per

coordinate over the whole track. The split with the smallest residual is accepted as a contact if it explains the pixels at least twice as well as the unsplit fit, which a full toss does not; otherwise no contact is declared. The contact instant is interpolated between the neighbouring frames as the time at which the two fitted arcs meet, and the contact pixel is the mean of the two arcs there.

D. Initialisation

The contact pixel back-projects to (x_b, y_b, R) . Seen from far down the pitch a pixel of error on the ground plane is metres, so three further candidates are tried: the ground crossings of six-parameter parabola fits to the pre- and post-contact frames, and the back-projections of the two frames adjacent to the split; the candidate whose refined fit explains the pixels best is kept. For each candidate, $\mathbf{v}^-$ and $\mathbf{v}^+$ come from the linear solves of Section III, and the initial restitution is $e_0 = -v_z^+/v_z^-$ clipped to $[0.25, 0.85]$.

E. Joint refinement

The delivery is parameterised by $\boldsymbol{\theta} = (t_b, x_b, y_b, v_x^-, v_y^-, v_z^-, v_x^+, v_y^+, v_z^+, e)$, with $v_z^+ = -e v_z^-$, so

$$\mathbf{p}(t) = (x_b, y_b, R) + \mathbf{v}^\mp \tau - \frac{1}{2} g \tau^2 \hat{\mathbf{z}}, \quad \tau = t - t_b, \quad (1)$$

taking $\mathbf{v}^-$ for $\tau < 0$ and $\mathbf{v}^+$ for $\tau \geq 0$. The residual vector stacks the weighted pixel residuals $w_i(u_i - \hat{u}_i)$ and $w_i(v_i - \hat{v}_i)$ of every frame with three soft priors: $(e - 0.55)/0.15$, $(v_x^+ - v_x^-)/4$ and $(v_y^+ - v_y^-)/2$, in units chosen so that each prior is worth about one pixel of disagreement. The post-contact horizontal velocity is free to differ from the pre-contact one, which is how seam and spin deviation and friction loss are measured; when the post-contact arc is only a few frames long the priors keep the problem determined and the covariance reports the resulting uncertainty. The cost is minimised with a soft-L1 loss [34] at a scale of 2 px by the bounded trust-region reflective method [35], [36], with t_b bounded to the track's span, e to $[0.25, 0.85]$, and velocities to physical ranges. The pixel radii are not used: they are unreliable from a box detector and the anchored model does not need them.

The covariance of the estimate is the linearised $\hat{\sigma}^2(J^\top J)^{-1}$ with J the Jacobian at the solution and $\hat{\sigma}^2$ the residual variance per degree of freedom [37]. It is a first-order approximation and Section VI-C measures how far it can be trusted.

F. Deliveries without an observed contact

A full toss, or a clip cut before the pitch, has no split. The six-parameter single parabola is then solved linearly [4] and refined with the apparent-size residual $w_i(r_i - fR/\hat{d}_i)$ at weight 0.3, the flight is expressed in the parameterisation of (1) by placing the unobserved contact where the parabola would meet the ground with the prior restitution, and the unobserved post-contact deviation and restitution are given their prior spread in the covariance. The result is flagged, and its uncertainty is large. This is also the estimator we compare against, since it is what a monocular method without the contact constraint can do.

G. Prediction and decision

The post-contact parabola is continued to the stump plane $x = 0$; if it would touch down again first, a further bounce with the same e is applied. The crossing (y^*, z^*) and its covariance follow from (1) and a numerical Jacobian of the crossing with respect to $\boldsymbol{\theta}$, giving σ_y and σ_z . The impact point is the model position at the last tracked frame.

Law 36 [28] is applied as three tests. The ball must not have pitched outside the line of the leg stump; it must have struck the batter in line with the stumps (impact outside the line of off is not out if a shot was offered, which we cannot see, so impact outside the line on either side is treated as not in line, the batter's benefit); and it must be projected to hit the stumps, meaning $|y^*| \leq w_o$ and $z^* \leq h_s$ for the centre. The playing conditions [29] make a centre within one ball radius outside a boundary umpire's call. We widen each band to $\max(R, k\sigma)$ with $k = 1$ and the relevant propagated sigma, so a decision the estimate cannot support at one sigma is returned as umpire's call rather than asserted. The choice of k trades false outs against umpire's calls and Section VI-D traces the curve.

V. EXPERIMENTAL SETUP

A. Synthetic deliveries with exact truth

A generator integrates each delivery at 1 kHz from release to the stump plane. Its physics is deliberately richer than the estimator's model: aerodynamic drag with $C_d = 0.5$ on a 0.16 kg ball, which at 40 m/s is a deceleration comparable to g [25]; a lateral acceleration before contact (swing [24]); a fractional loss of horizontal speed and an added lateral velocity at contact (friction and turn); and a restitution coefficient drawn per delivery from 0.40 to 0.70, the range of real pitches [27]. Release speed is drawn from 60 to 150 km/h, pitching length from 1 to 10 m, line from -0.6 to 0.6 m, release height from 1.8 to 2.3 m. The batter's pad intercepts the ball at the popping crease, 1.22 m from the stumps, so detections stop there and the estimator must predict the last 1.22 m; the true stump-plane crossing is the integrated trajectory continued untouched. The true verdict is computed from the true contact, crease and stump-plane points by the same Law 36 tests with $k = 0$, so the three classes, out, not out and umpire's call, are exactly the classes a perfect instrument would return.

Observations are the projections of the true positions at the frame rate, with Gaussian pixel noise on the centre, 10% multiplicative noise on the radius, a dropout probability per frame, and the calibration taps perturbed by Gaussian noise before the pose is re-solved, so the estimator never receives the true camera. The reference configuration is 60 Hz, 1 px centre noise, 5% dropout, 2 px tap noise, all physical effects on, and a phone 2.8 m behind the striker's stumps at 1.7 m height. Each factor is swept in turn over 400 random deliveries per setting, and six camera placements are tested at the reference noise: three heights at the striker's end, a lateral offset of 1.5 m, and two placements at the bowler's end. Both estimators, anchored and single parabola, receive identical detections.

B. Rendered clips through the full pipeline

To include detection and association error, the fixed 100-delivery grid of five speeds (90 to 145 km/h), five lines ($0, \pm 0.2, \pm 0.4$ m) and four lengths (2, 4, 6 and 8 m) is rendered to 1080 by 1920 portrait clips at 60 Hz with a flat pitch, stumps, creases and a motion-smeared ball, and each clip is submitted to the production job with the calibration marks perturbed by 2 px, exactly as the app submits it. The learned detector was not trained on rendered frames, so this tests the geometry end to end rather than the detector's robustness to clutter, which the real clips do.

C. Real clips

Three hand-held phone clips of net deliveries, an outdoor off-spinner with a white ball and two indoor deliveries with a red ball, are analysed with their stump and corner taps. No metric truth exists for them, so two things are reported that need none: the output with its stated uncertainty, and repeatability. Each clip is analysed on all sampled frames, on the even frames only and on the odd frames only; the two half-rate analyses see disjoint detections of the same delivery, and the spread between their stump-plane predictions is an empirical precision to hold against the propagated sigma.

D. Metrics

Verdict agreement is the fraction of deliveries whose returned class equals the true class. A false out is a not-out delivery returned as out, the error a review system must not make; a missed out is the reverse. Errors are absolute differences between the predicted and true stump-plane crossing in y and z , and between the estimated and true contact point; medians and 90th percentiles are reported because the distributions are heavy-tailed. Coverage is the fraction of deliveries whose absolute error lies within one, two and three propagated sigmas.

VI. RESULTS

A. Reconstruction under controlled error

Table I and Fig. 2 give the controlled result. At the reference configuration the anchored estimator agrees with the true verdict in 95.2% of deliveries with a median stump-plane error of 2.4 cm laterally and 3.8 cm vertically, and 90th percentiles of 9.2 and 11.0 cm. The single parabola on the same detections agrees in 30.2% with median errors of 27.8 and 43.5 cm. The factor of ten is the observability argument of Section III made numerical: without the anchor the depth-and-speed direction is unresolved, and a lateral error of a quarter of a metre at the stumps is a coin toss on a target 26 cm wide.

Across the sweeps the anchored estimator's agreement lies between 88.0% and 97.5%, and the single parabola's between 26.0% and 36.8%. Three features of the sweeps matter. Frame rate has almost no effect from 30 to 240 Hz (94.2% at 30 Hz), because the anchor is a single well-determined event and the parabolas on either side are over-determined from a handful of frames; a 30 Hz phone is enough. Calibration tap error has little effect until it is large (95.5% at 8 px, 91.0% at 12 px),

because the pose is solved from twelve marks and the stump plane, not any single tap, sets the anchor. Dropped frames to 40% cost nothing measurable (96.0%). What does cost is pixel noise on the ball centre: agreement is 97.5% at zero noise, 95.0% at 2 px, 93.0% at 3 px and 88.0% at 5 px, with the vertical error rising from 2.4 to 8.0 cm, and the response is the intended one: the number of umpire's calls rises from 38 to 72 while false outs stay at 1 or fewer in every row. The unmodelled physics, drag, swing, friction and spin together, moves agreement from 97.0% to 95.5% and the median errors from 1.1 and 2.2 cm to 2.8 and 3.9 cm: the constant-velocity model absorbs drag into an average speed over the short post-contact arc, and the free post-contact velocity absorbs friction and turn.

Table II pools every anchored reconstruction. Of 9110 deliveries that were not out, 6 were returned as out (0.07%), and of 512 that were out, 3 were returned as not out. Nearly every disagreement is between a decisive class and umpire's call in one direction or the other, and the decisive-for-marginal cases lie within a few millimetres of a band edge. The median release-speed error is 14.4 km/h and the contact point is located to 4.7 cm along the pitch and 1.1 cm across it, so release speed and pitching point, which a monocular estimator without the anchor cannot state, are measurements here.

B. Camera placement

Table III gives the placement result. Any position behind the striker's stumps works: low, high or back, agreement is 94.0% to 95.5%, and a 1.5 m offset to the side gives 90.5%, with median errors of a few centimetres in every case. The bowler's end is worse, 88.0% and 86.0%, with median vertical errors of 11.2 and 9.7 cm, and the reason is geometric rather than incidental: from behind the bowler the decisive post-contact arc is 15 to 20 m away and foreshortened, a pixel of contact error on the ground plane is a metre along the pitch, and the multiple-candidate initialisation of Section IV is what holds the failure rate at 7 of 400; a single-candidate initialisation lost dozens more. The practical instruction is simple and the app gives it: film from behind the batter.

C. How far the stated uncertainty can be trusted

Fig. 3 tests the covariance. The one-sigma bound covers 50.5% of lateral and 60.6% of vertical errors, the two-sigma bound 79.3% and 84.7%, and the three-sigma bound 91.4% and 93.0%, against 68%, 95% and 99.7% for a Gaussian. The median ratio of error to sigma is 0.99 laterally and 0.76 vertically. The stated uncertainty is therefore slightly optimistic, with heavier tails than a linearised covariance predicts, as expected of a first-order approximation of a model with a bounded parameter and priors; it is why the decision bands are widened by $k\sigma$ rather than replaced by it. A sigma from this system should be read as roughly a 55% bound, not a 68% one; the median sigma at the stump plane is 2.6 cm laterally and 5.3 cm vertically.

TABLE I

EXPERIMENT 1. BOTH ESTIMATORS ON IDENTICAL SYNTHETIC DETECTIONS, 400 DELIVERIES PER ROW, ONE FACTOR VARIED FROM THE REFERENCE CONFIGURATION (60 HZ, 1 PX, 2 PX TAPS, 5% DROPOUT, PHYSICS ON). ERRORS ARE AT THE STUMP-PLANE CROSSING; FALSE OUT COUNTS NOT-OUT DELIVERIES RETURNED AS OUT.

Factor	Value	Anchored (this paper)				Single parabola			
		Agree (%)	False out	Median y/z (cm)	90th y/z (cm)	Agree (%)	False out	Median y/z (cm)	90th y/z (cm)
Pixel noise σ (px)	0	97.5	0	1.9 / 2.4	7.6 / 8.1	31.2	6	30.5 / 47.0	327.2 / 93.8
	0.5	96.8	0	2.1 / 2.8	6.6 / 8.6	29.5	3	26.9 / 46.0	297.6 / 99.7
	1	95.2	0	2.4 / 3.8	9.2 / 11.0	30.2	8	27.8 / 43.5	350.1 / 101.8
	2	95.0	0	2.9 / 4.5	10.4 / 13.6	32.2	3	29.9 / 48.0	353.9 / 101.5
	3	93.0	0	4.0 / 6.2	13.4 / 17.8	31.2	0	28.6 / 40.6	289.6 / 100.4
	5	88.0	1	5.3 / 8.0	17.6 / 29.2	27.0	5	24.5 / 47.4	497.7 / 104.6
Frame rate (Hz)	30	94.2	1	2.8 / 4.1	9.3 / 11.8	26.0	2	23.2 / 39.3	315.8 / 93.7
	60	96.2	0	2.3 / 3.4	8.4 / 11.7	30.5	6	29.6 / 46.0	384.1 / 94.4
	120	96.2	0	2.4 / 3.1	8.9 / 11.1	34.2	3	31.9 / 44.6	356.2 / 89.7
	240	95.0	1	2.1 / 3.2	7.9 / 11.2	35.0	6	30.4 / 37.9	417.0 / 94.6
Tap noise σ (px)	0	97.2	0	2.3 / 3.5	7.4 / 9.8	32.0	5	27.8 / 41.2	373.5 / 94.5
	2	94.8	0	2.1 / 3.4	8.5 / 10.0	32.0	3	20.1 / 36.6	230.4 / 96.9
	4	93.8	0	2.5 / 3.9	9.7 / 11.7	26.0	3	26.2 / 40.7	347.2 / 97.9
	8	95.5	0	2.8 / 4.4	9.5 / 12.8	34.5	3	23.9 / 40.3	222.5 / 100.3
	12	91.0	0	3.2 / 5.4	11.1 / 17.3	29.2	4	32.8 / 42.2	305.6 / 98.1
Dropped frames	0	95.8	0	2.2 / 3.2	8.5 / 9.9	32.2	3	26.1 / 39.1	229.1 / 93.8
	0.1	96.2	0	2.4 / 3.7	8.3 / 11.4	32.5	4	27.7 / 36.7	327.8 / 94.3
	0.25	96.5	0	2.3 / 4.0	8.7 / 11.3	30.2	3	30.3 / 46.0	266.7 / 94.9
	0.4	96.0	0	2.4 / 3.7	9.6 / 11.3	26.2	3	25.7 / 43.0	333.3 / 98.5
Unmodelled physics	off	97.0	0	1.1 / 2.2	3.5 / 6.6	36.8	2	12.7 / 34.9	332.2 / 105.2
	on	95.5	0	2.8 / 3.9	8.2 / 11.3	30.5	3	30.6 / 33.8	331.9 / 87.4

TABLE II

EXPERIMENT 1, ALL 10800 ANCHORED RECONSTRUCTIONS POOLED. ROWS ARE THE TRUE CLASS, COLUMNS THE RETURNED CLASS.

Truth \ returned	Not out	Out	Umpire's call	No verdict
Not out (9110)	8872	6	208	24
Out (512)	3	365	144	0
Umpire's call (1178)	166	63	943	6

D. The decision band

Table IV traces what k buys. With the Laws' fixed band alone ($k = 0$) the system returns 13 false outs and 12 missed outs over 10800 deliveries and umpire's call on 10.4%; at $k = 1$ the false outs fall to 6 and the umpire's calls rise to 12.0%; at $k = 2$ there are 4 false outs and 14.3% umpire's calls. The curve is the operating choice a deployment makes. For coaching, where a wrong decisive call mis-teaches and an umpire's call invites a look at the replay, we ship $k = 1$; officiating, which this system is not for, would want the false-out rate at $k = 2$ and would decline one delivery in 7.

E. End to end on rendered clips

Through the production job, with the learned detector, RANSAC association and the calibration re-solved from perturbed taps, the pipeline agrees with the true verdict on 86 of 100 deliveries with 0 false outs and 0 missed outs; every other disagreement is an umpire's call (Table V). The median stump-plane error is 2.4 cm laterally and 8.7 cm vertically, the contact point is located to 7.7 cm along the pitch and 0.6 cm across it, and the release speed to 8.6 km/h. The vertical error is larger than in Experiment 1 because the detector's centroid on a motion-smeared ball is biased along the streak, which is a pixel-noise level nearer 2 to 3 px than the 1 px of the reference

configuration, and the propagated sigma reflects it: the median stated sigma is 5.0 cm laterally and 18.5 cm vertically, and the system answers with umpire's call on 15 deliveries.

The structure of the disagreements is physical. Yorkers at 2 m agree in 76% of cases with 6 umpire's calls, because the ball is intercepted 0.8 m after it pitches and the post-contact arc holds one or two frames; at 4, 6 and 8 m agreement is 92, 84 and 92%. On the middle-stump line, where 15 of the 20 deliveries are truly out, 11 are returned as umpire's call: the vertical band at the top of the stumps is where the propagated sigma is widest, and the system declines rather than asserts. That is a conservative operating point, and Table IV says what a smaller k would recover.

F. Real footage

Table VI reports the real clips without embellishment. The outdoor off-spinner (reproj. 10.6 px, 29 tracked frames) is reconstructed with an observed contact at 7.7 m and a release speed of 66 km/h. The all-frames analysis returns not out, placing the impact outside the off line; the two half-rate analyses return umpire's call. Their stump-plane predictions differ by 21 cm and 26 cm, inside the stated sigma, but that sigma is 49 cm laterally and 147 cm vertically on the full track and larger on the halves: with a calibration residual the app itself flags, this clip supports little, and the output says so. The indoor leg-spinner is returned not out on the pitching test in all three analyses, with the contact consistently 26 to 29 cm outside the leg line, but the half-rate stump-plane predictions differ by 69 cm laterally against a stated 16 cm on the full track: on this clip that sigma is not honest. The reason is visible in the table, the two halves found the contact at different distances and each states a sigma of over a metre, so the linearised covariance of the full-track solution does

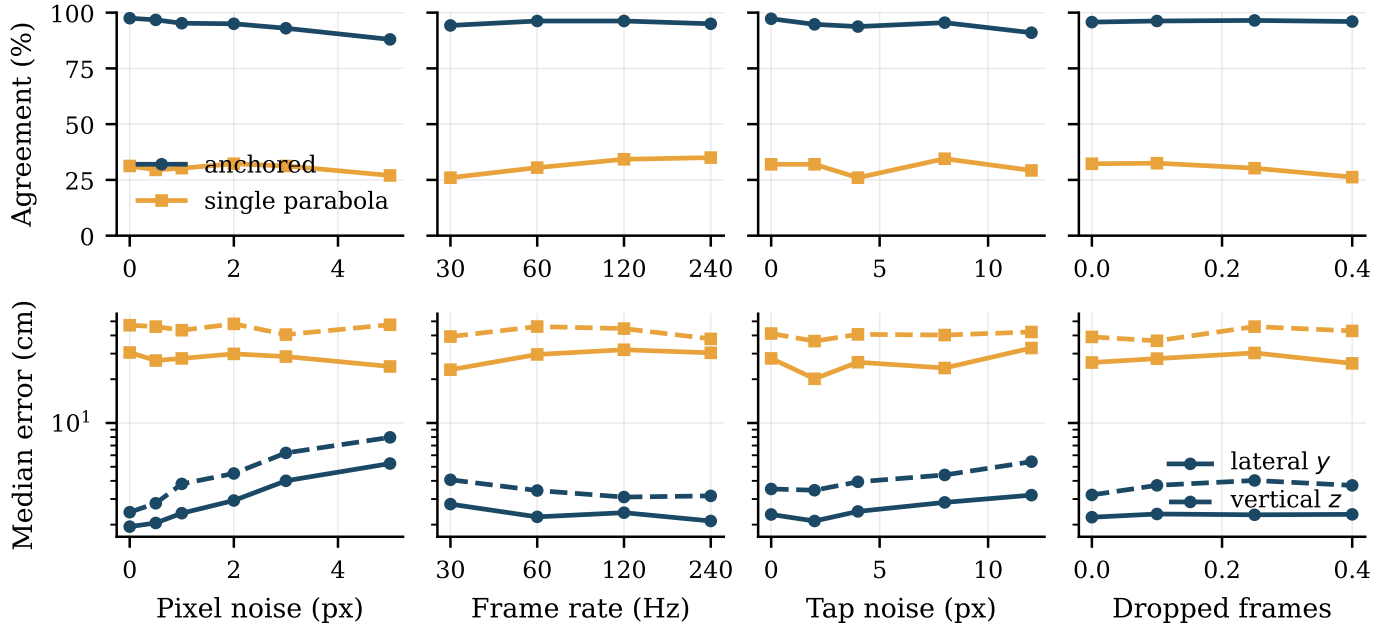


Fig. 2. Experiment 1. Top: verdict agreement against each measurement factor. Bottom: median stump-plane error, lateral (solid) and vertical (dashed), for the anchored estimator (blue) and the single parabola (orange). The anchored estimator is flat in every factor until pixel noise reaches several pixels.

TABLE III
EXPERIMENT 1, SIX PHONE PLACEMENTS AT THE REFERENCE NOISE, ANCHORED ESTIMATOR.

Camera	Agree (%)	False out	Umpire's call	No verdict	Median y/z (cm)	90th y/z (cm)
Striker's end, low (1.2 m)	95.5	1	47	1	2.7 / 2.3	9.0 / 7.8
Striker's end, reference (1.7 m)	94.5	0	53	1	2.5 / 3.4	9.2 / 10.9
Striker's end, high and back (2.6 m, 4 m)	94.0	1	57	0	2.4 / 5.3	9.0 / 14.7
Striker's end, offset 1.5 m	90.5	0	55	0	4.4 / 2.8	11.5 / 8.0
Bowler's end, reference	88.0	1	45	7	4.9 / 11.2	16.0 / 27.6
Bowler's end, offset 2 m, high	86.0	0	60	1	7.3 / 9.7	19.1 / 19.7

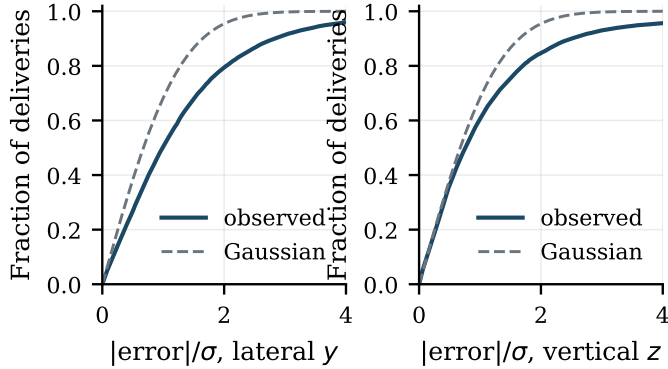


Fig. 3. Experiment 1. Empirical distribution of the absolute error divided by the propagated sigma, over every anchored reconstruction with an observed contact, against the Gaussian expectation.

not see the alternative that half of the frames supports. The third clip, a short indoor net whose fitted length is 4.4 m, is declined: the anchored fit leaves a residual of 19 px against an acceptance threshold of 12, and the system returns no verdict rather than one it cannot defend. Real footage carries the errors the synthetic study does not model, rolling-shutter distortion, a moving camera, occlusion by the batter and a net rather than

TABLE IV
EXPERIMENT 1, ALL ANCHORED RECONSTRUCTIONS, VERDICT AS A FUNCTION OF THE BAND WIDENING k . DECISIVE IS THE FRACTION RETURNED AS OUT OR NOT OUT.

k	Agree (%)	False out	Missed out	Umpire's call (%)	Decisive (%)
0	94.5	13	12	10.4	89.3
0.5	94.4	10	6	11.1	88.6
1	94.3	6	3	12.0	87.7
1.5	93.9	5	3	13.0	86.7
2	93.2	4	2	14.3	85.4

a pitch, and on these three clips the honest summary is that the system's declining and its umpire's calls are working as designed, and its stated uncertainty is not yet a guarantee.

VII. LIMITATIONS

No metric truth on real footage. Every accuracy number comes from synthetic deliveries, rendered or not; the real clips show only self-consistency. The repeatability result makes the next step necessary: real deliveries with instrumented truth, most simply a second synchronised phone at right angles, so the covariance can be checked in the field.

The uncertainty is a first-order approximation. Coverage is below the Gaussian nominal at every level, and one real clip

TABLE V
EXPERIMENT 2. THE FULL PIPELINE, DETECTOR INCLUDED, ON THE RENDERED 100-DELIVERY GRID, BY FACTOR.

Factor	Value	n	Agree (%)	Umpire's call	Median y (cm)	Median z (cm)
Length (m)	2	25	76	6	2.2	14.4
	4	25	92	2	1.2	2.7
	6	25	84	4	5.1	15.5
	8	25	92	3	4.0	8.4
Speed (km/h)	90	20	90	3	0.6	2.1
	105	20	80	4	2.2	5.0
	120	20	75	5	3.9	11.1
	135	20	95	1	3.0	14.8
	145	20	90	2	6.2	14.4
Line (m)	-0.4	20	100	0	5.5	10.2
	-0.2	20	95	1	4.1	10.0
	+0.0	20	35	14	0.4	10.7
	+0.2	20	100	0	3.9	8.2
	+0.4	20	100	0	4.0	7.6

TABLE VI
EXPERIMENT 3. THREE HAND-HELD NET CLIPS, EACH ANALYSED ON ALL SAMPLED FRAMES AND ON THE EVEN AND ODD FRAMES ALONE. ERRORS ARE NOT MEASURABLE; THE COLUMNS REPORT WHAT THE SYSTEM STATES AND HOW IT VARIES.

Clip	Frames	Reproj. (px)	Track	Contact x (m)	Stump plane $y \pm \sigma$ (cm)	Stump plane $z \pm \sigma$ (cm)	Verdict
Outdoor net, white ball, off-spin	all	10.6	29	7.7	$+41 \pm 49$	119 ± 147	not out
	even	10.6	13	7.2	$+49 \pm 102$	92 ± 179	umpire's call
	odd	10.6	15	7.5	$+71 \pm 705$	65 ± 338	umpire's call
Indoor net, red ball, leg-spin	all	3.7	37	4.9	-39 ± 16	44 ± 177	not out
	even	3.7	17	2.8	-12 ± 376	56 ± 213	not out
	odd	3.7	17	4.8	-81 ± 161	104 ± 247	not out
Indoor net, red ball, off-spin	all	2.0	36	—	—	—	declined
	even	2.0	31	—	—	—	declined
	odd	2.0	36	—	—	—	declined

shows why: when the contact split is ambiguous the solution is multimodal and a linearised covariance about one mode does not know about the other. A sampling-based uncertainty or an explicit test of the alternative split would address this.

The estimator ignores drag, swing and spin. The constant-velocity model absorbs them over the short arcs it fits, at a cost of 97.0% to 95.5% agreement, but a long pre-contact arc under strong swing would be fitted worse, and the release speed reported is an average over the observed arc, not the speed at the hand.

The contact must be seen. A full toss, a clip cut before the pitch, or a camera position from which the batter hides the contact falls back to the single-parabola estimator, whose accuracy is the orange curve in Fig. 2, and whose result is flagged as such.

Calibration is manual. Twelve taps on the stumps and pitch corners are required. Detecting the stumps and creases automatically would remove the step and the tap error with it.

VIII. CONCLUSION

A single phone behind the batter can reconstruct a cricket delivery in metres if it uses the one event the delivery always contains. The bounce fixes a point on a known plane at a known instant, and with that anchor the monocular projectile problem, ill-conditioned along the depth of a ball flying toward the camera, becomes two over-determined linear problems joined by a restitution coefficient. On 10800 synthetic deliveries with exact truth the anchored estimator agrees with the

true verdict in 94.3% of cases with 6 false outs and median stump-plane errors of 2.7 and 3.9 cm, against 33.6% and errors ten times larger without the anchor; through the full pipeline it agrees on 86 of 100 with no false out. The propagated uncertainty is slightly optimistic and is reported as such, and it is used as the Laws' umpire's call is used: to decline a call the evidence does not support. The system is a coaching instrument that says how sure it is, not a substitute for a calibrated multi-camera rig, and the paper has tried to state where the gap lies.

DATA AND CODE AVAILABILITY

The pipeline, the synthetic generator and renderer, the experiment scripts, the raw result files and the scripts that generate every number, table and figure in this manuscript are available at <https://github.com/kafel/pocket-drs> under the AGPL-3.0 licence. A checker, `paper/experiments/verify_claims.py`, regenerates the numbers from the raw files and fails if the manuscript is stale.

DECLARATIONS

The author declares no competing interests and received no funding. The real clips were recorded by the author at net sessions with the consent of those present and contain no identifiable personal data at the resolution used.

REFERENCES

- [1] N. Owens, C. Harris, and C. Stennett, "Hawk-eye tennis system," in *Proc. Int. Conf. Visual Information Engineering (VIE)*, 2003, pp. 182–185.
- [2] H. Collins and R. Evans, "You cannot be serious! Public understanding of technology with special reference to "Hawk-Eye"," *Public Understanding of Science*, vol. 17, no. 3, pp. 283–308, 2008.
- [3] B. Bal and G. Dureja, "Hawk eye: a logical innovative technology use in sports for effective decision making," *Sport Science Review*, vol. 21, no. 1–2, pp. 107–119, 2012.
- [4] E. Ribnick, S. Atev, and N. P. Papanikolopoulos, "Estimating 3D positions and velocities of projectiles from monocular views," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 31, no. 5, pp. 938–944, 2009.
- [5] G. Van Zandycke and C. De Vleeschouwer, "3D ball localization from a single calibrated image," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022, pp. 3471–3479.
- [6] S. Shirmohammadi and A. Ferrero, "Camera as the instrument: the rising trend of vision based measurement," *IEEE Instrumentation & Measurement Magazine*, vol. 17, no. 3, pp. 41–47, 2014.
- [7] Z. Zhang, D. Xu, and M. Tan, "Visual measurement and prediction of ball trajectory for table tennis robot," *IEEE Transactions on Instrumentation and Measurement*, vol. 59, no. 12, pp. 3195–3205, 2010.
- [8] S. Wang, Y. Xu, Y. Zheng, M. Zhu, H. Yao, and Z. Xiao, "Tracking a golf ball with high-speed stereo vision system," *IEEE Transactions on Instrumentation and Measurement*, vol. 68, no. 8, pp. 2742–2754, 2019.
- [9] G. Di Leo, C. Liguori, and A. Paolillo, "Covariance propagation for the uncertainty estimation in stereo vision," *IEEE Transactions on Instrumentation and Measurement*, vol. 60, no. 5, pp. 1664–1673, 2011.
- [10] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2004.
- [11] G. Thomas, R. Gade, T. B. Moeslund, P. Carr, and A. Hilton, "Computer vision for sports: current applications and research topics," *Computer Vision and Image Understanding*, vol. 159, pp. 3–18, 2017.
- [12] P. R. Kamble, A. G. Keskar, and K. M. Bhurchandi, "Ball tracking in sports: a survey," *Artificial Intelligence Review*, vol. 52, no. 3, pp. 1655–1705, 2019.
- [13] H.-T. Chen, M.-C. Tien, Y.-W. Chen, W.-J. Tsai, and S.-Y. Lee, "Physics-based ball tracking and 3D trajectory reconstruction with applications to shooting location estimation in basketball video," *Journal of Visual Communication and Image Representation*, vol. 20, no. 3, pp. 204–216, 2009.
- [14] V. Chao, A. Jamsrandorj, Y. M. Oo, K.-R. Mun, and J. Kim, "3D ball trajectory reconstruction of a ballistic shot from a monocular basketball video," in *Proc. 49th Annu. Conf. IEEE Ind. Electron. Soc. (IECON)*, 2023, pp. 1–6.
- [15] P. Ponglertnapakorn and S. Suwajanakorn, "Where is the ball: 3D ball trajectory estimation from 2D monocular tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2025, pp. 6112–6121.
- [16] B. Ghimire, G. K. Mishra, and B. R. Lamichhane, "3D ball trajectory from a single camera: a cost-effective ball estimation technique for cricket," *TechRxiv preprint*, 2026.
- [17] P. Liu and J.-H. Wang, "MonoTrack: shuttle trajectory reconstruction from monocular badminton video," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022, pp. 3512–3521.
- [18] J. Calandre, R. Péteri, L. Mascarrilla, and B. Tremblais, "Extraction and analysis of 3D kinematic parameters of table tennis ball from a single camera," in *Proc. Int. Conf. Pattern Recognition (ICPR)*, 2021, pp. 9468–9475.
- [19] R. Voeikov, N. Falaleev, and R. Baikulov, "TTNet: real-time temporal and spatial video analysis of table tennis," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 3866–3874.
- [20] Y.-C. Huang, I.-N. Liao, C.-H. Chen, T.-U. Ik, and W.-C. Peng, "TrackNet: a deep learning network for tracking high-speed and tiny objects in sports applications," in *Proc. IEEE Int. Conf. Advanced Video and Signal Based Surveillance (AVSS)*, 2019, pp. 1–8.
- [21] N.-E. Sun, Y.-C. Lin, S.-P. Chuang, T.-H. Hsu, D.-R. Yu, H.-Y. Chung, and T.-U. Ik, "TrackNetV2: efficient shuttlecock tracking network," in *Proc. Int. Conf. Pervasive Artificial Intelligence (ICPAI)*, 2020, pp. 86–91.
- [22] Y.-J. Chen and Y.-S. Wang, "TrackNetV3: enhancing shuttlecock tracking with augmentations and trajectory rectification," in *Proc. ACM Multimedia Asia*, 2023, pp. 1–7.
- [23] F. Yan, W. Christmas, and J. Kittler, "Layered data association using graph-theoretic formulation with applications to tennis ball tracking in monocular sequences," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1814–1830, 2008.
- [24] R. D. Mehta, "An overview of cricket ball swing," *Sports Engineering*, vol. 8, no. 4, pp. 181–192, 2005.
- [25] G. Robinson and I. Robinson, "The motion of an arbitrarily rotating spherical projectile and its application to ball games," *Physica Scripta*, vol. 88, no. 1, p. 018101, 2013.
- [26] M. J. Carré, D. M. James, and S. J. Haake, "Impact of a non-homogeneous sphere on a rigid surface," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, vol. 218, no. 3, pp. 273–281, 2004.
- [27] D. M. James, M. J. Carré, and S. J. Haake, "The playing performance of county cricket pitches," *Sports Engineering*, vol. 7, no. 1, pp. 1–14, 2004.
- [28] Marylebone Cricket Club, "The laws of cricket, 2017 code (3rd edition, 2022)," London, U.K.: MCC, 2022, law 36, Leg before wicket.
- [29] International Cricket Council, "ICC men's test match playing conditions, appendix D: Decision review system," Dubai, U.A.E.: ICC, 2023.
- [30] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.
- [31] V. Lepetit, F. Moreno-Noguer, and P. Fua, "EPnP: an accurate O(n) solution to the PnP problem," *International Journal of Computer Vision*, vol. 81, no. 2, pp. 155–166, 2009.
- [32] G. Terzakis and M. Lourakis, "A consistently fast and globally optimal solution to the perspective-n-point problem," in *Computer Vision – ECCV 2020*, ser. Lecture Notes in Computer Science, vol. 12346, 2020, pp. 478–494.
- [33] M. A. Fischler and R. C. Bolles, "Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.
- [34] B. Triggs, P. F. McLauchlan, R. I. Hartley, and A. W. Fitzgibbon, "Bundle adjustment – a modern synthesis," in *Vision Algorithms: Theory and Practice*, ser. Lecture Notes in Computer Science, vol. 1883, 2000, pp. 298–372.
- [35] M. A. Branch, T. F. Coleman, and Y. Li, "A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems," *SIAM Journal on Scientific Computing*, vol. 21, no. 1, pp. 1–23, 1999.
- [36] P. Virtanen, R. Gommers, T. E. Oliphant *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.
- [37] G. A. F. Seber and C. J. Wild, *Nonlinear Regression*. New York: Wiley, 1989.
- [38] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun, "Towards robust monocular depth estimation: mixing datasets for zero-shot cross-dataset transfer," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 3, pp. 1623–1637, 2022.
- [39] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 4104–4113.
- [40] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [41] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," <https://github.com/ultralytics/ultralytics>, 2023, software, version 8.